

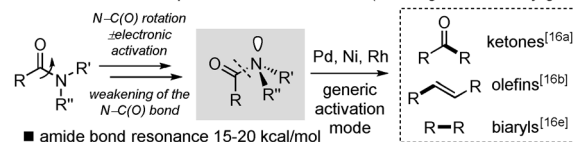
Cross-Coupling

Efficient Synthesis of Diaryl Ketones by Nickel-Catalyzed Negishi Cross-Coupling of Amides by Carbon–Nitrogen Bond Cleavage at Room Temperature Accelerated by a Solvent Effect

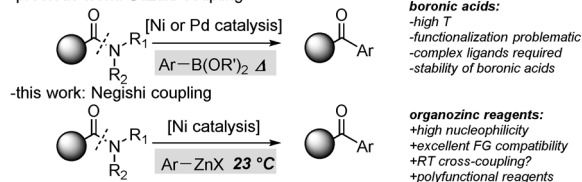
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Abstract: The first Negishi cross-coupling of amides for the synthesis of versatile diaryl ketones by selective C–N bond activation under exceedingly mild conditions is reported. The cross-coupling was accomplished with bench-stable, inexpensive precatalyst $[\text{Ni}(\text{PPh}_3)_2\text{Cl}_2]$ that shows high functional-group tolerance and enables the synthesis of highly functionalized diaryl ketone motifs. The coupling occurred with excellent chemoselectivity favoring the ketone (cf. biaryl) products. Notably, this process represents the mildest conditions for amide N–C bond activation accomplished to date (room temperature, < 10 min). Considering the versatile role of polyfunctional biaryl ketone linchpins in modern organic synthesis, availability, and excellent functional-group tolerance of organozinc reagents, this strategy provides a new platform for amide N–C bond/organozinc cross-coupling under mild conditions.

A. Amide activation concept for N–C metal insertion (challenge: amide conjugation)

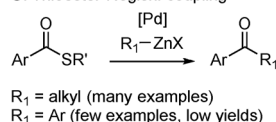


B. Diaryl ketone synthesis from amides by N–C activation: current state-of-the-art -previous work: Suzuki coupling



the mildest amide N–C cross-coupling reported to date

C. Thioester Negishi coupling



D. Diaryl ketones

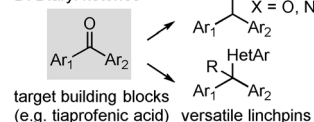


Figure 1. A) Amide bond activation concept for metal catalysis. B) Examples of amide N–C bond cross-coupling. C) Classic acyl electrophiles in Negishi reaction. D) Diaryl ketone linchpins.

Diaryl ketones are important structural motifs in a wide array of pharmaceuticals, biologically active natural products and functional organic materials.^[1] Moreover, diaryl ketones serve as multifaceted linchpin intermediates that can be readily converted into a variety of fundamental diarylmethane building blocks of particular significance in organic synthesis,^[2] such as bisbenzylic amines, alcohols, and heterocycles. Transition-metal catalyzed acylation of organometallics with acyl electrophiles, arguably, represents the most important method for the synthesis of diaryl ketones.^[3] In 2004, Knochel and co-workers documented the iron-catalyzed arylation of acyl cyanides with functionalized arylmagnesium reagents.^[4] While focused on magnesium nucleophiles,^[5] this pioneering study forecast the attractive possibility that complex diaryl ketones might be directly prepared from functionalized organometallic reagents.^[5,6]

The Negishi cross-coupling reaction represents a powerful transition-metal catalyzed coupling manifold, whereby the in-

trinsic high transmetalation activity and the exceptional functional-group tolerance of organozinc reagents provides a direct route to polyfunctionalized building blocks.^[6] Tremendous recent progress was achieved by Knochel and co-workers through the development of polyfunctionalized organozinc reagents and their application in a wide range of cross-couplings.^[7] The development of new catalytic systems by Organ and co-workers^[8] and mechanistic studies by the group of Lei are noteworthy.^[9] Elegant examples of the synthesis of biaryl motifs by direct cross-coupling of aryl chlorides,^[10a,b] aryl pivalates,^[10c] and aryl ammonium salts^[10d] with organozinc reagents have been reported. Recently, Stambuli demonstrated the first diaryl ketone synthesis from less stable thioester electrophiles and organozinc reagents;^[11] however, it is worthwhile to note that, unlike alkylation, diaryl ketone synthesis using organozinc reagents presents a formidable challenge due to incompatibility of oxidative addition/transmetalation steps,^[12] which results in low yields (Figure 1). Most notably, despite these significant advances,^[6–12] simple amides have yet to be employed as electrophilic coupling partners in Negishi reactions. This constitutes a surprising gap in the organozinc cross-coupling methods in light of: 1) the high bench stability of amides; 2) the

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ease of preparation; 3) the fact that amides are typically inert to the reaction conditions allowing for ring-prefunctionalization; and, most importantly, 4) the prevalence of amides in biology (peptides, proteins) and as key structural features in pharmaceuticals, drug candidates, and agrochemicals, which could allow for a strategic disconnection to access a myriad of biologically active molecules.^[13]

The major challenge in employing amides as electrophilic coupling partners in Negishi cross-coupling by N–C cleavage is the low reactivity for direct oxidative addition^[13] (amide bond resonance of 15–20 kcal mol^{−1}, $n_N \rightarrow \pi^*_{CO}$ conjugation) and control of the stability of the resulting acyl metal intermediate.^[14] In 2015, we have introduced the concept of steric activation of amide bonds to enable new valuable C–C bond-forming reactions to occur directly from amides^[15] in a series of transition-metal catalyzed reactions.^[16a–e] Metal insertion into the amide N–C bond can proceed only if the amide bond (ca. 40% double bond character) has been distorted from planarity by steric and/or electronic means.^[15] Independently, Garg and Zou reported Ni- and Pd-catalyzed Suzuki cross-coupling reactions of acyclic twisted imides.^[16f–h] Altogether, these complementary methods demonstrate that typically inert amide bonds can be activated towards unconventional reactivity, thus providing a new powerful method of amide disconnection in organic synthesis.

Herein, we advance this concept to report the first Negishi cross-coupling of amides for the synthesis of versatile diaryl ketones by selective C–N bond activation^[17] under exceedingly mild conditions. The reaction is based on the use of inexpensive, bench-stable Ni precatalysts that are economically advantageous over Pd catalysts.^[18] The reaction proceeds with excellent ketone (cf. biaryl)^[16e] chemoselectivity and represents the mildest conditions for amide N–C bond activation accomplished to date (room temperature, < 10 min).^[16, 17] Considering the unique features of this coupling manifold, the importance of biaryl ketone linchpins,^[1, 2] and excellent functional-group tolerance of organozinc reagents,^[6, 7] this method provides a new platform for amide N–C bond/organozinc cross-coupling.^[6, 7]

Given our interest in developing new generic strategies for activation of inert N–C amide bonds,^[16a–e] we recently questioned whether amides might be exploited as unique, bench-stable substrates in Negishi cross-coupling reactions to afford diaryl ketones under mild conditions. At the outset, it was unclear whether such a scenario would be feasible considering: 1) the high activation barrier for metal insertion into the N–C bond, typically requiring harsh conditions;^[16] 2) few precedents for the coupling of aryl (cf. alkyl) zinc reagents with acyl electrophiles;^[12] and 3) control of the relative reactivity of acyl (cf. aryl) cross-coupling of amide electrophiles.^[14]

We started our investigations by evaluating the cross-coupling of amides **1** with arylzinc reagents under a variety of conditions (Figure 2).^[16] Screening was performed with PhZnCl, and efforts were made to develop operationally simple reaction conditions. After significant experimentation, we found that the proposed Negishi reaction was indeed feasible using the sterically distorted amide **1d** with the Ni^{II}-based precatalyst

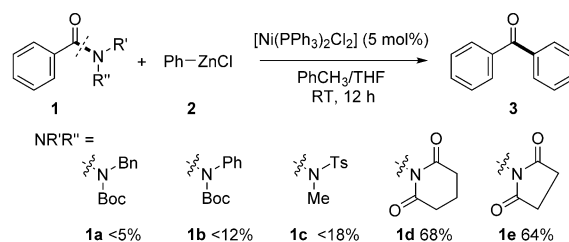


Figure 2. Ni-catalyzed Negishi cross-coupling by N–C amide bond cleavage: the effect of N-substituents. Conditions: **1** (1.0 equiv), PhZnCl (1.5 equiv), [Ni(PPh₃)₂Cl₂] (5 mol %), solvent (0.20 M), 23 °C.

system.^[18] Interestingly, using the less distorted amides **1b**, **1c**^[15] a promising yield of the diaryl ketone could be obtained, demonstrating the high reactivity of the catalyst system. Moreover, the marginally less distorted amide **1e**^[15] afforded the desired product in good yield. These initial observations demonstrate the viability of the challenging acyl–aryl amide Negishi coupling^[11, 12] from a range of electronically and sterically diverse amides. Importantly, decarbonylation of the acyl–metal intermediate was not observed.^[14] It is worth noting that the insertion occurred selectively into the amide

Table 1. Optimization of Ni-catalyzed Negishi diaryl ketone synthesis by coupling of amides by N–C cleavage.^[a]

Entry	Catalyst	L	Solvent	Yield [%] ^[b]
1	[Pd(PPh ₃) ₄]	–	THF	34
2	[Pd(PPh ₃) ₄]	–	toluene	62
3	[PdCl ₂ (PPh ₃) ₂]	–	toluene	60
4	[PdCl ₂ (PPh ₃) ₂]	–	dioxane	36
5	[PdCl ₂ (PPh ₃) ₂]	DMF ^[d]	toluene	54
6	[PdCl ₂ (PPh ₃) ₂]	LiCl ^[e]	toluene	65
7	[PdCl ₂ (PPh ₃) ₂]	CuI ^[f]	toluene	< 5
8	[Pd(acac) ₂]	–	dioxane	13
9	[Pd ₂ (dba) ₃]	–	dioxane	23
10	Pd(OAc) ₂	–	dioxane	10
11	[PdCl ₂ (CH ₃ CN) ₂]	–	toluene	8
12	[NiCl ₂ (dppe)]	–	toluene	29
13	[NiCl ₂ (dppf)]	–	toluene	48
14	Ni(OAc) ₂ ·4 H ₂ O	–	toluene	56
15	[Ni(PCy ₃) ₂ Cl ₂]	–	toluene	17
16	[Ni(acac) ₂]	–	toluene	62
17	[NiCl ₂ (PPh ₃) ₂]	–	toluene	80
18	[NiCl ₂ (PPh ₃) ₂]	PPh ₃	toluene	20
19	[NiCl ₂ (PPh ₃) ₂]	–	THF	62
20	[NiCl ₂ (PPh ₃) ₂]	–	DMF	48
21	[NiCl ₂ (PPh ₃) ₂]	–	Et ₂ O	92 ^[c]
22	–	–	toluene	< 2

[a] See the Supporting Information for full experimental details. Conditions: amide (0.1 mmol), R = CF₃, Ph–ZnCl (1.5 equiv), catalyst (5 mol %), solvent (0.20 M), 23 °C, 12 h. acac = acetylacetonate; dba = dibenzylideneacetone; dppe = 1,2-bis(diphenylphosphino)ethane; dppf = 1,1'-bis(diphenylphosphino)ferrocene. [b] GC/¹H NMR yields. [c] Isolated yield. [d] 2 equiv. [e] 5 equiv. [f] 1 equiv. see the Supporting Information for details.

N–C(O) bond and scission of the alternative σ N–C amide bond was not observed.^[19,20]

Table 1 summarizes key optimization results in the development of the amide Negishi cross-coupling reaction. The optimization was performed using 4-trifluoromethylphenyl amide as the coupling partner. Various catalysts were tested, and $[\text{Ni}(\text{PPh}_3)_2\text{Cl}_2]$ showed the best catalytic activity (Table 1, entries 1–17). Interestingly, the nature of the ligand on Ni only moderately affects the yield, with several Ni^{II} precatalysts affording the desired product in good yields (entries 12–17). This finding is consistent with the ionic character of the C–Zn bond and high nucleophilicity of organozinc reagents.^[6c] Additional ligands had a deleterious effect (entry 18). Moreover, Pd catalysts serve as viable promoters for the reaction (entries 1–11). This represents the first example of Pd-catalyzed Negishi coupling of amides,^[11,12] thus providing an important entry point for future optimization. Interestingly, polar solvents were not required for the reaction, which represents a significant practical

advantage.^[21] Various organozinc reagents were tested,^[6,7] with ArZnCl providing the best results. The classic studies by the Organ group on the salt effect in the transmetalation of zinc reagents highlighted the crucial role of halide salt additives in the coupling of ArZnX reagents in nonpolar solvents.^[21] Extensive screening of various solvents revealed that the use of diethyl ether gave the best results (entries 19–21, see the Supporting Information). We hypothesize that this cosolvent might be acting to stabilize the putative dihalo-bridged aryl–zinc intermediate.^[9b] The discovery that the use of diethyl ether facilitates challenging Negishi couplings may find applications in other Negishi aryl cross-couplings of acyl electrophiles.^[6,12] Consistent with our hypothesis, the reaction was not observed in the absence of Ni catalysts (entry 22).

With the optimized procedure in hand, the scope of this new Negishi reaction variant was then evaluated (Table 2). A wide variety of functionalized aryl amide electrophiles with a diverse set of substitution could be accommodated, including 3-

Table 2. Ni-catalyzed Negishi biaryl ketone synthesis by cross-coupling of amides with aryl zinc halides.^[a,b]

 (3a) 84%	 (3b) 79%	 (3c) 57%	 (3d) 76%	 (3e) 66%
 (3f) 76%	 (3g) 71%	 (3h) 56%	 (3i) 80%	 (3j) 82%
 (3k) 52%	 (3l) 92%	 (3m) 92%	 (3n) 84%	 (3o) 78%
 (3p) 80%	 (3q) 85%	 (3r) 90%	 (3s) 72%	 (3t) 85% ^[c]
 (3u) 88%	 (3v) 51%	 (3w) 53%	 (3x) 72%	 (3y) 97%
 (3z) 90%	 (3aa) 66%	 (3ab) 96%	 (3ac) 76%	 (3ad) 71%
 (3ae) 74%	 (3af) 84%	 (3ag) 64%	 (3ah) 75%	 (3ai) 67%

[a] Amide (1.0 equiv), R–ZnCl (1.5 equiv), $[\text{Ni}(\text{PPh}_3)_2\text{Cl}_2]$ (5 mol %), THF/Et₂O (0.20 M), 23 °C. [b] Isolated yields. [c] Gram scale. See the Supporting Information for details.

fluoro, *ortho*-methyl, 3-thienyl, 3-trifluoromethyl, 2-thienyl, 4-methoxy, 4-methyl, 4-chloro-, 2-furyl, 3,4-difluoro-, 3-methoxy, and 1,3-dioxolane substituents (**3a–o**), affording the desired benzophenone products in good to excellent yields. Interestingly, variation of the electronic nature of the aromatic ring had little impact on the reaction efficiency, with neutral **3m**, highly electron-withdrawing **3i**, and electron-donating substituents **3o** providing the desired ketone products in high yields. This reactivity compares very favorably with the related Suzuki coupling by amide N–C cleavage,^[16c,f] which tends to be sensitive to the electronics of the electrophile. The reaction could be further extended to the coupling of functionalized aryl zinc nucleophiles (**3p–ah**), affording the desired benzophenones in good to excellent yields. Of particular note are the cross-couplings of aryl amide electrophiles with indolyl (**3t–w**), dimethylamino (**3ag,ah**), and substituted *ortho*-methoxy aryl zinc nucleophiles (**3x–ac**). These reactions furnish functionalized indolyl, amino- and methoxy-substituted aryl ketones that possess a wide range of biological activities.^[1e–g] Note that these *ortho*-methoxy-substituted nucleophiles are not tolerated in Suzuki amide N–C couplings.^[16c,f] The current limitation is that 4-ethoxycarbonylphenyl- and 4-cyanophenyl-zinc reagents, readily prepared by direct halide/magnesium exchange,^[7] are not tolerated in the reaction; however, the scope of the reaction surpasses the Suzuki cross-coupling of amides. Pharmaceutically relevant heterocycles, such as thiophene, furane, dioxolane, and indole are perfectly accommodated. The Negishi coupling proceeds under ambient conditions (cf. Suzuki), which is important for practical applications.^[22] Additional studies to expand the scope of this reaction are underway in our laboratories.

Importantly, the cross-coupling was carried out on a gram scale in excellent yield (**3s**). Given its user-friendly nature (room temperature, bench-top), economically viable profile of components (cf. boronic acids, Pd), the method provides an entry point for the utilization of amide Negishi coupling in industrial development.^[22]

A striking feature of this new Negishi cross-coupling reaction is the rate of acyl–aryl coupling under exceedingly mild conditions. Monitoring of the reaction profile of **31** indicated that the reaction is complete in less than 10 min at room temperature. This marks the fastest N–C amide bond cross-coupling achieved to date,^[16,17] which bodes well for the development of cross-coupling processes with functionalized organozinc reagents by amide N–C bond activation.

In conclusion, we have demonstrated the first diaryl ketone synthesis by the Negishi cross-coupling of amides by chemoselective N–C activation. This versatile method is distinguished by a broad substrate scope, functional-group tolerance and mild reaction conditions unattainable by the current state-of-the-art Suzuki couplings with amide electrophiles. Notably, the method represents the mildest conditions for amide N–C activation accomplished to date. Given the importance of polyfunctional biaryl motifs and high functional-group tolerance of organozinc reagents, these studies lay a foundation for a general application of amide Negishi cross-coupling for the synthesis of polyfunctionalized ketones.

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Keywords: diaryl ketones • N–C activation • Negishi coupling • nickel • zinc

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Cross-Coupling

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Efficient Synthesis of Diaryl Ketones by Nickel-Catalyzed Negishi Cross-Coupling of Amides by Carbon–Nitrogen Bond Cleavage at Room Temperature Accelerated by a Solvent Effect



Easy crossing: The first Negishi cross-coupling of amides for the synthesis of versatile diaryl ketones by selective C–N bond activation under exceedingly mild conditions is reported (see scheme). The cross-coupling was accomplished

with bench-stable, inexpensive $[Ni^{II}(PPh_3)_2Cl_2]$ precatalyst that shows high functional-group tolerance and enables the synthesis of highly functionalized diaryl ketone motifs.